VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE ADVANCED - 4 | PAPER - 2

JEE 2022

Date	Maximam Marks	Duration
10th October, 2021	198	3 Hours

PAPER SCHEME

1. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III: **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).

2. **Section 1** contains **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.

   Section 2 contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

   Section 3 contains **6 Numerical Value Type Questions**. The answer to each of the question is a numerical value. For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places.** ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25)**

3. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ____

Roll Number : ____

OMR Bar Code Number : ____

Candidate's Signature : __________________ Invigilator's Signature ______________

MARKING SCHEME

SECTION – 1 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE**.
  For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks :	+3 If ONLY the correct integer is entered;
Zero Marks :	0 If the question is unanswered:
Negative Marks:	–1 In all other cases.

SECTION - 2 (Maximum Marks: 24)

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If only (all) the correct option(s) is(are) chosen
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark: 0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks: –2 In all other cases.

SECTION – 3 (Maximum Marks: 24)

- This section has **SIX (06)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. **If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)**
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 In all other cases.

SUBJECT I : PHYSICS 66 MARKS

SECTION 1
SINGLE DIGIT INTEGER

This section consists of SIX (06) questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

1. A series circuit consists of a resistance $R = 100\Omega$, a coil of inductance $L = 2\text{ mH}$, a capacitor of capacitance $C = 10\ \mu\text{F}$ and an AC source of angular frequency $\omega = 200\text{ rad/s}$. If the rms value of the potential difference across the capacitor is 40 V, the rms value of the potential difference (in Volts) across the resistance is _____________.

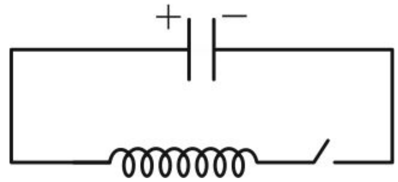
2. N identical uncharged capacitors are connected in series with a battery and a switch. The switch is closed and kept closed until the capacitors are completely charged. Let the total potential energy stored now in all the capacitors be U_I. Now, the switch is opened, the battery is removed from the circuit and an identical uncharged capacitor is connected in its place. Now the switch is closed again and kept closed for a long time. The total potential energy stored in all the capacitors is now U_F. If $\frac{U_F}{U_I} = \frac{1}{6}$, then N is ________________.

3. A metal conductor is placed in a room that is maintained at a constant temperature $T_0 = 20^\circ C$ and a potential difference is applied across the conductor. The conductor loses heat from its surface at a rate $\frac{dQ}{dt} = k(T - T_0)$, where k is a constant, T is the temperature of the conductor in $^\circ C$ and $\frac{dQ}{dt}$ is in Watts. Assume that the conductor has high thermal conductivity, so its entire volume is at the same temperature at any instant. When a constant potential difference 50 V is applied across the conductor, after a long time the current through it achieves a steady state value of 6A and its temperature becomes $120^\circ\ C$. The value of the constant k is ___________ $\text{W}/^\circ\text{C}$.

4. Two identical conducting plates of surface area A each are fixed parallel to each other with a small separation d between them. The plates are given total charge $+3Q$ and $-Q$. The potential difference between the plates is $X\left(\frac{Qd}{\varepsilon_0 A}\right)$. The value of X is ___________.

5. A small non-conducting ball carrying positive charge 0.01 C hangs in equilibrium from an ideal spring of spring constant 100 N/m as shown. Now, a vertically downward electric field of magnitude 2000 V/m is switched on. In its subsequent motion, the maximum kinetic energy (in Joules) of the ball is _____________.

6. A capacitor of capacitance $10\ \mu F$ is charged to a potential difference 60 V and connected to a coil of inductance 4 mH as shown. The resistance in the circuit is negligible. After the switch is closed, the maximum value that the current in the circuit reaches is ____________ A.



SPACE FOR ROUGH WORK

SECTION 2
MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), **(C)** and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. A particle of mass m carrying positive charge of magnitude q moves in a circular path in a uniform magnetic field of intensity B. If the kinetic energy of the particle is K, the magnetic moment produced by it:

(A) has a magnitude $\frac{K}{B}$

(B) has a magnitude $\frac{K}{2B}$

(C) is in the same direction as the magnetic field

(D) is in the opposite direction to the magnetic field

8. Three particles carrying charge of equal magnitude are fixed at the vertices of an equilateral triangle ABC. The electric field at the centroid G of the triangle is directed along $\overrightarrow{GB}$. Which of these options is/are correct?

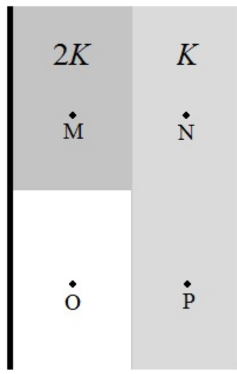
(A) The particles at A and C carry positive charge and the particle at B carries negative charge

(B) If the particle at C is moved to a very large distance from the other particles, the electric field at G becomes $\frac{\sqrt{3}}{2}$ times

(C) If (starting from the initial situation) the particle at B is moved to G, the electric field at B is directed along $\overrightarrow{GB}$

(D) If (starting from the initial situation), the particle at A is released, its initial acceleration is parallel to $\overrightarrow{CB}$

9. Two dielectric slabs of dielectric constant K and $2K$ are placed inside a parallel-plate capacitor as shown. The separation between the plates is d. The thickness of both slabs is $\frac{d}{2}$. The slab of dielectric constant $2K$ covers half the area of the left plate, while the slab of dielectric constant K covers the entire area of the right plate. The remaining space is vacuum. The capacitor is now charged such that the potential difference between the plates is ΔV. The magnitude electric field at the points M, N, O and P is E_M, E_N, E_O and E_P. Which of the following statements is/are correct?



(A) $E_P = \frac{E_O}{K}$ **(B)** $E_M = \left(\frac{K+1}{3K}\right)E_O$ **(C)** $E_N = \frac{E_M}{2}$ **(D)** $\frac{\Delta V}{d} = \left(\frac{K+1}{2}\right)E_O$

10. A thin metallic rod XY of length L is pivoted at a point M on the rod which is at a distance $\frac{L}{4}$ from end X. If the rod is rotated at a constant angular velocity ω about an axis passing through M and perpendicular to the rod in the presence of a uniform magnetic field of magnitude B directed perpendicular to the plane of rotation of the rod, then:

(A) The emf induced between M and Y is $\frac{9}{32}B\omega L^2$

(B) The emf induced between M and Y is $\frac{9}{16}B\omega L^2$

(C) The emf induced between X and Y is $\frac{1}{8}B\omega L^2$

(D) The emf induced between X and Y is $\frac{1}{4}B\omega L^2$

11. Two straight long wires fixed in the X-Y plane along the lines $x = -a$ and $x = a$ carry steady current of magnitude I in the +Y and –Y directions respectively. Let the magnetic field at a point $P(0,0,z)$ on the positive Z-axis, and at a point $Q(0,0,-z)$ on the negative Z-axis be $\vec{B}_P$ and $\vec{B}_Q$ respectively. Which of the following statements is/are correct?

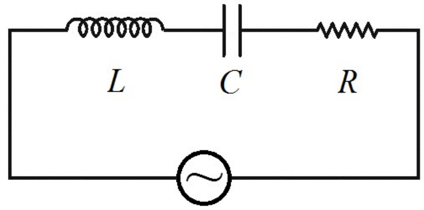
(A) $\vec{B}_P$ and $\vec{B}_Q$ are respectively along the –Z and +Z directions

(B) $\vec{B}_P$ and $\vec{B}_Q$ are both along the –Z direction

(C) $|\vec{B}_P|$ and $|\vec{B}_Q|$ are both proportional to $\frac{1}{z^2}$ if $z \gg a$

(D) $|\vec{B}_P|$ and $|\vec{B}_Q|$ are both proportional to $\frac{1}{z^3}$ if $z \gg a$

12. A series LCR circuit is connected to a source of alternating voltage of angular frequency ω. Which of the following statements is/are **TRUE** if $\omega = \frac{1}{\sqrt{LC}}$ and **FALSE** if $\omega \neq \frac{1}{\sqrt{LC}}$?



(A) The capacitor is uncharged at the instant the current in the circuit reaches its maximum value
(B) The rms potential difference across the capacitor is equal to the rms EMF induced across the inductor
(C) The rate of heat dissipation is maximum at the instant the current is maximum
(D) The combined energy stored in the capacitor and the inductor remains constant

SPACE FOR ROUGH WORK

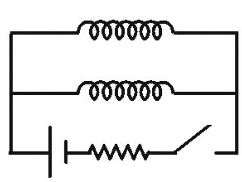
SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. ***In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*****. (Example: 6, 81, 1.50, 3.25)**

13. A charge particle of mass $m = 1.8\ kg$ and charge $q = 1C$ is projected on a rough horizontal *x-y* plane. Electric field and magnetic field given by $\vec{E} = -2\left(\frac{N}{C}\right)\hat{k}$ and $\vec{B} = -\frac{1}{2}(T)\hat{k}$ respectively exist in the region. The charge enters the region at $(4m, 0, 0)$ with velocity $\vec{v} = 5\left(\frac{m}{s}\right)\hat{j}$. Coefficient of friction between the particle and plane is 0.1. Radius of curvature of path of particle 1s after projection is ________ meters. Take $g = 10\frac{m}{s^2}$

14. Two coils of self-inductance $L_1 = 1$ mH and $L_2 = 2$ mH and negligible internal resistance are connected in parallel across a battery. The mutual inductance of the coils is negligible. Initially, the switch is open and current through both coils is zero. A long time after the switch is closed, the ratio of the current in the coils, $\frac{I_1}{I_2}$ is equal to _____________.



15. A superconducting ring of radius $R = 1m$ and self inductance L is kept coaxial with a long solenoid of radius $r = 0.5m$. If current in the solenoid is decreased to $\frac{i_0}{3}$ from i_0, where i_0 is initial current in the solenoid. Current induced in the ring is $\frac{i_0}{k}$. Value of k is ________.

$\left[\frac{L}{n} = \pi\mu_0, \text{ where } n \text{ is number of turns/length in the solenoid}\right]$

16. A thin, insulating annular disc of inner and outer radius $\frac{R}{2}$ and R is charged uniformly over its surface with a total charge Q. The disc is rotated about an axis passing through its centre and perpendicular to its plane at a constant angular velocity ω. The magnetic moment of the disc is $X(QR^2\omega)$. The value of X is ____________.

17. Suppose we have four uncharged capacitors, two of capacitance $4\ \mu\text{F}$ and two of capacitance $2\ \mu\text{F}$. Each capacitor can only bear a maximum potential difference 100 V without getting damaged. The maximum capacitance $(\text{in } \mu\text{F})$ that can be achieved by a combination of all four capacitors such that a potential difference 200 V can be applied across the combination without damaging any of the capacitors is ______________.

18. A coil of self-inductance 40 mH and internal resistance $9\ \Omega$ is used as a heater. It is connected to a source of alternating voltage of amplitude 300 V and angular frequency 300 rad/s and used to heat 2 kg of water placed in a vessel of heat capacity $3000\ \text{J/}^\circ\text{C}$. The heat capacity of the coil is $1200\ \text{J/}^\circ\text{C}$. Initially, the coil, the water and the vessel, all are at temperature $20\ ^\circ\text{C}$. Assuming that no heat is lost to the surroundings, and that the coil, the water and the vessel always remain at the same temperature, the time (in seconds) after which the temperature of the water has risen to $60\ ^\circ\text{C}$ is ____________.

(Specific heat of water $= 4200\ \text{J/kg}^\circ\text{C}$)

SPACE FOR ROUGH WORK

SUBJECT II : CHEMISTRY 66 MARKS

SECTION 1

SINGLE DIGIT INTEGER

This section consists of SIX (06) questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

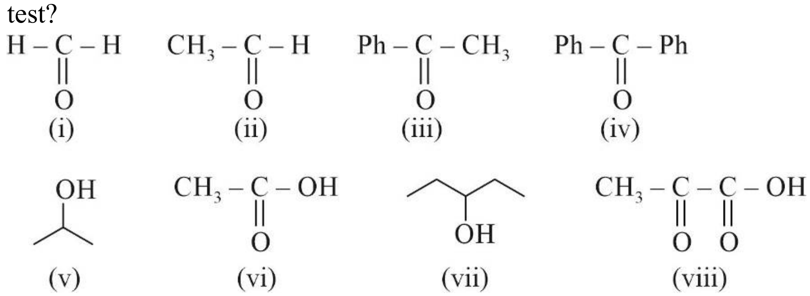
1. For the reaction: $2NO(g) + H_2(g) \longrightarrow N_2O(g) + H_2O(g)$ the value of $\frac{-dp}{dt}$ was found to be 1.5 Pa sec^{-1} for a pressure of 372 Pa for NO and 0.25 Pa sec^{-1} for a pressure of 152 Pa for NO, the pressure of H_2 being constant. If pressure of NO was kept constant, the value of $-\frac{dp}{dt}$ was found to be 1.6 Pa sec^{-1} for a pressure of 288 Pa for H_2 and 0.80 Pa sec^{-1} for a pressure of 144 Pa for H_2. If the order of reaction with respect to NO and H_2 are a and b, respectively then the value of $(a + 2b)$ is ______.
$[\log_{10} 2 = 0.3,\ \log_{10} 3 = 0.5,\ \log_{10} (2.45) = 0.4]$

2. A metal 'M' (atomic mass = 31.25 amu) crystallizes in ccp but it has some vacancy defect. If the edge length of the unit cell is 500 pm and the density of the metal is 1.6075 g/cm^3, then the number of moles of metal atoms missing per litre of the crystal is __________ . (1 amu = 1.67×10^{-24} g)

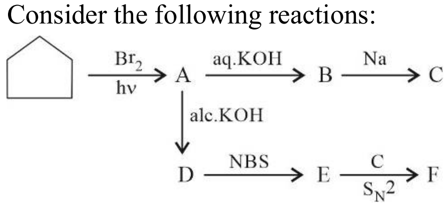
3. A quantity of 1.04 g of $CoCl_3.6NH_3$ (molar mass = 267 g/mol) was dissolved in 100g of H_2O. The freezing point of the solution was –0.29°C. The number of NH_3 molecules in coordination sphere of octahedral complex $CoCl_3.6NH_3$ is___________ . [K_f for water = 1.86°C/m]

4. By passing a certain amount of charge through NaCl solution, 9.08L of chlorine gas were liberated at STP. When the same amount of charge is passed through a nitrate solution of metal M, 52.8 g of the metal was deposited. If the atomic weight of metal is 200 amu, then valency of the metal is _______ .

5. Consider following compounds and find the number of compounds which give positive iodoform test?



6. Consider the following reactions:

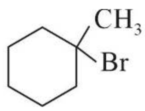


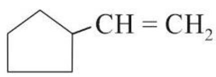
The degree of unsaturation of organic product 'F' is _________ .

SECTION 2
MULTIPLE CORRECT ANSWERS TYPE

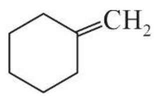
This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. In the given reaction

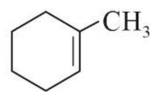
C_7H_{12} (P) $\xrightarrow{HBr}$  ; P can be:



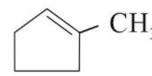
(A)



(B)



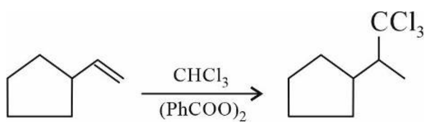
(C)

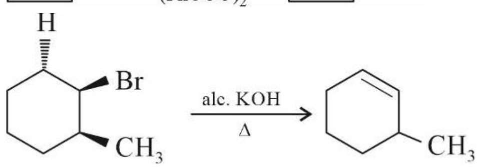


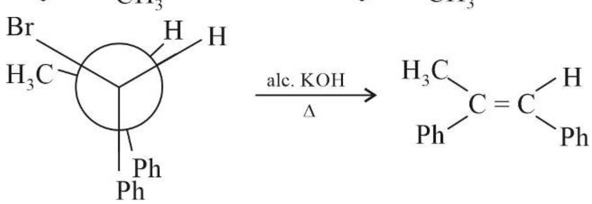
(D)

8. Which of the following reaction given is/are correctly showing major product:

(A) $H_3C-CH=CH_2 \xrightarrow[H_2O_2]{HBr} H_3C-CH_2-CH_2-Br$

(B) 

(C) 

(D) 

9. Which of the following statement(s) is/are incorrect regarding the defects in solids:

(A) AgCl crystal shows both Schottky and Frenkel defect

(B) Impurity defect by doping of Arsenic in silicon results 'n' type semiconductor

(C) Doping in crystal introduces dislocation defect

(D) Metal deficiency defect can occur only with extra anion present in the interstitial voids

10. In a study of effect of temperature on rate of reaction, the value of $\frac{1}{k}.\frac{dk}{dT}$ is found to be $\frac{1.25\times10^6}{T^3}K^{-1}$, where k is rate constant of reaction. Identify the correct statement(s):

(A) The activation energy for the reaction at 250K is 10 kcal/mol

(B) The activation energy for the reaction at 2000K is 1.25 kcal/mol

(C) The rate of increase of rate constant with increase in temperature is higher at lower temperature than at higher temperature

(D) The value of $\frac{d(\ell nk)}{dT}$ is 0.625 at 1000K

11. For the coagulation of a particular colloidal solution, the order of coagulation power of some electrolytes is in the order: $Na_3PO_4 > BaSO_4 > AlCl_3$. Which of the following information(s) may be correct for such colloid?

(A) In electro osmosis, the dispersion medium move towards anode

(B) Diffused layer around colloidal particles in the electrical double layer mostly contains negatively charged particles

(C) It may be basic dye such as methylene blue

(D) It may be metal sol

12. Two litre solution of a buffer mixture containing 1M NaH_2PO_4 and 1M Na_2HPO_4 is placed in two compartments (one litre in each) of an electrolytic cell. The platinum electrodes are inserted in each compartment and 1.25A current is passed for 965min. Assuming electrolysis of only water at each compartment, what will be pH in each compartment after passage of above charge?

(pKa for $H_2PO_4^- = 2.15$, $\log 7 = 0.85$, $\log 2 = 0.3$, $\log 3 = 0.48$)

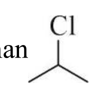
(A) Anode: 3.00 **(B)** Cathode: 3.00 **(C)** Anode: 1.30 **(D)** Cathode: 1.30

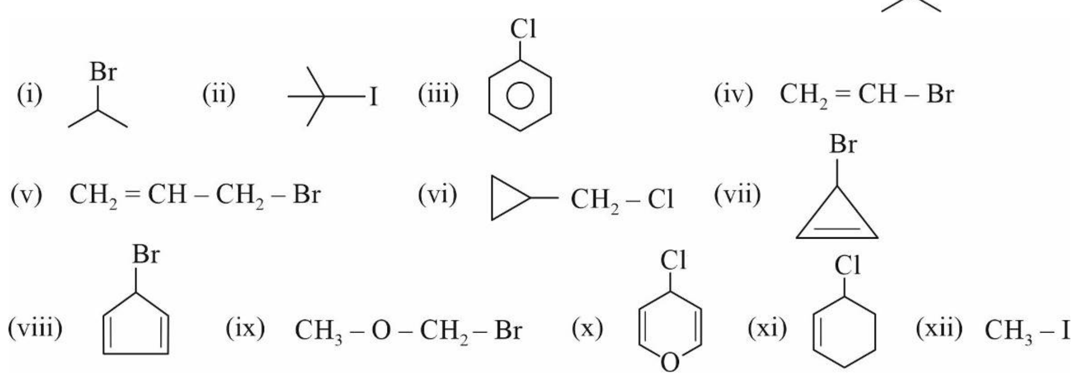
SPACE FOR ROUGH WORK

SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

13. How many compounds among the following can show faster S_N1 reaction than 



14. Consider the following reaction:

$$CHCl_3 \xrightarrow[\Delta]{Ag} (A) \xrightarrow{\text{Red hot Iron tube}} (B) \xrightarrow{Cl_2, Fe} (C) \xrightarrow[\text{Conc.}H_2SO_4]{\text{Chloral,}} (D) \xrightarrow{NBS} (E)$$

Sum of number of halogen atoms present in products 'D' and 'E' is ____________ .

15. Consider the following reactions:

$$\text{1-Bromobutane} \xrightarrow[\text{dry ether}]{\text{Mg}} \text{A} \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CH_3-CH_2-CHO} \text{B} \xrightarrow[\Delta]{\text{Conc. } H_2SO_4} \text{C (Major)}$$

$$\text{C} \xrightarrow{Br_2/CCl_4} \text{D} \xrightarrow{\text{fused KOH followed by } NaNH_2} \text{E}$$

$$\text{F} \xleftarrow[HgSO_4,\ H_2SO_4]{H_2O} \text{E}$$

What is the molar mass of compound 'F' in g mol^{-1}.
[Molar mass of C = 12, H = 1, O = 16, N = 14, Br = 80 g mol^{-1}]

16. The conductivity of saturated solution of sparingly soluble salt, $Ba_3(PO_4)_2$ is $1.2 \times 10^{-5}\ \text{ohm}^{-1}\ \text{cm}^{-1}$. The limiting equivalent conductivity of $BaCl_2, K_3PO_4, KCl$ are 160, 140, 100 ohm^{-1} cm^2 eq^{-1} respectively. If K_{sp} of $Ba_3(PO_4)_2$ is $x \times 10^{-25}$ then the value of x is _______ .

17. When cells of the skeletal vacuole of a frog were placed in a series of NaCl solutions of different concentrations at 6°C, it was observed microscopically that they remain unchanged in x% NaCl solution, shrank in more concentrated solution and swells in more dilute solutions. Water freezes from the x% salt solution at –0.4°C. Then osmotic pressure of the cell cytoplasm at 6°C is ________ atm. (Assume molarity equal to molality) (K_f of water = 1.86°K/m, R = 0.08 $\frac{\text{atm}-\text{L}}{\text{mol}-\text{K}}$)

18. Ammonia gas adsorbs at 4g charcoal having rough surface area of about $10^{16}\,\text{m}^2$ and area occupied by each molecule of NH_3 is $\frac{10^{-3}}{18}\text{cm}^2$. The mass (in g) of ammonia adsorbed per gram of charcoal is______. $\left(N_A = 6 \times 10^{23}, \text{ Molar mass of H} = 1,\ \text{N} = 14\,\text{g mol}^{-1}\right)$

SPACE FOR ROUGH WORK

SUBJECT III : MATHEMATICS 66 MARKS

SECTION 1

SINGLE DIGIT INTEGER

This section consists of SIX (06) questions. The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 TO 9. **BOTH INCLUSIVE.**

1. Let $F(x)$ be a non-negative continuous function defined on R such that $F(x)+F\left(x+\frac{1}{2}\right)=3$ and the value of $\int_0^{1500} F(x)\,dx$ is $\frac{9000}{\lambda}$. Then the numerical value of λ is _________.

2. If $\int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x+\cos^8 x}dx = k\left(\frac{\pi}{2}\right)^2$, then the value of k is _______.

3. If $x\frac{dy}{dx}=x^2+y-2,\ y(1)=1$, then $y(2)$ equals _____.

4. Let $f:[1,\infty)$ be a differentiable function such that $f(1)=2$. If $6\int_1^x f(t)\,dt = 3x\,f(x)-x^3$ for all $x\ge 1$, then the value of $f(2)$ is _____.

5. Let $\overrightarrow{OA}=\vec{a}, \overrightarrow{OB}=2\vec{a}+10\vec{b}, \overrightarrow{OC}=\vec{b}$ where O, A, C are non collinear points. Let λ denote the area of the quadrilateral $OABC$. Let m denotes the area of parallelogram with $\overrightarrow{OA}$ and $\overrightarrow{OC}$ as adjacent sides. If $\lambda = 2\lambda m$ then find the value of λ. $(\lambda \ne 0)$

6. If $\hat{i}\times\left[(\vec{a}-\hat{j})\times\hat{i}\right]+\hat{j}\times\left[(\vec{a}-\hat{k})\times\hat{j}\right]+\hat{k}\times\left[(\vec{a}-\hat{i})\times\hat{k}\right]=0$

and $\vec{a}=x\hat{i}+y\hat{j}+z\hat{k}$ then $8(x^3-xy+zx)$ is equal to :

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SECTION 2
MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

7. Let $A(k)$ be the area bounded by the curves $y = x^2 - 3$ and $y = kx + 2$

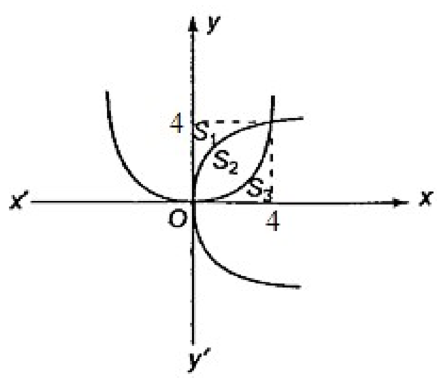
(A) The range of $A(k)$ is $\left[\frac{10\sqrt{5}}{3}, \infty\right)$

(B) The range of $A(k)$ is $\left[\frac{20\sqrt{5}}{3}, \infty\right)$

(C) If function $k \to A(k)$ is defined for $k \in [-2, \infty)$, then $A(k)$ is many-one function

(D) The value of k for which area is minimum is 1

8. The parabolas $y^2 = 4x$ and $x^2 = 4y$ divide the square region bounded by the lines $x = 4,\ y = 4$ and the co-ordinate axes. If S_1, S_2, S_3 are the areas of these parts numbered from top to bottom, respectively, then:



(A) $S_1 : S_2 = 1:1$ **(B)** $S_2 : S_3 = 1:2$ **(C)** $S_1 : S_3 = 1:1$ **(D)** $S_1 : (S_1 + S_2) = 1:2$

9. In which of the following differential equations degree is not defined?

(A) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x \log \frac{d^2y}{dx^2}$ **(B)** $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{d^2y}{dx^2}\right)$

(C) $x = \sin\left(\frac{dy}{dx} - 2y\right), |x| < 1$ **(D)** $x - 2y = \log\left(\frac{dy}{dx}\right)$

10. Which of the following statements is/are correct?

(A) If $\vec{n} \cdot \vec{a} = 0, \vec{n} \cdot \vec{b} = 0$ and $\vec{n} \cdot \vec{c} = 0$ for some non-zero vector $\vec{n}$, then $[\vec{a}\ \vec{b}\ \vec{c}] = 0$

(B) There exists a vector making angles 30º and 45º with x and y axes respectively

(C) Locus of point for which $x = 3$ and $y = 4$ is a line parallel to the z-axis whose distance from the z-axis, is 5

(D) The vertices of a regular tetrahedron are O, A, B, C where 'O' is the origin. The vector $\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$ is perpendicular the plane ABC.

11. If $\vec{a}=\hat{i}+2\hat{j}+4\hat{k}, \vec{b}=2\hat{i}-3\hat{j}+\hat{k}, \vec{c}=\hat{i}+4\hat{j}-4\hat{k}$, then the vector $\vec{a}\times(\vec{b}\times\vec{c})$ is orthogonal to:

(A) $\vec{a}$ **(B)** $\vec{b}$ **(C)** $\vec{c}$ **(D)** $\vec{a}+\vec{b}+\vec{c}$

12. Tangents are drawn to circle $x^2+y^2=32$ from a point A lying on x-axis. The tangent cuts y-axis at point B & C then coordinates of A such that area of ΔABC is minimum may be :

(A) (8, 0) **(B)** (6, 0) **(C)** $(-8, 0)$ **(D)** $(-6, 0)$

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SECTION 3

NUMERICAL VALUE TYPE QUESTIONS

This section has SIX (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25)

13. If the area bounded by the curve $f(x)=x^{1/3}(x-1)$ and x-axis is A, then the value of A is ______.

14. If the area enclosed by the curve $y=\sqrt{x}$ and $x=-\sqrt{y}$, the circle $x^2+y^2=2$ above the x-axis, is A then the value of $\frac{A}{\pi}$ is ______.

15. If the solution of the differential equation $\frac{dy}{dx}-y=1-e^{-x}$ and $y(0)=y_0$ has a finite value, when $x\to\infty$, then the value of y_0 is ______.

16. The real value of m for which the substitution $y=u^m$ will transform the differential equation $2x^4y\frac{dy}{dx}+y^4=4x^6$ in to a homogeneous equation is:

17. $y=f(x)$ passes through (1, 2) and satisfies the relation $y(1+xy)dx-xdy=0$. Find value of $7f\left(\frac{1}{2}\right)$.

18. The value of $\int\frac{(x^2-1)dx}{x^3\sqrt{2x^4-2x^2+1}}$ is $\frac{1}{2}\sqrt{2-\frac{l}{x^m}+\frac{1}{x^n}}+c$, $(l>0)$. The value of $l-m-n$ is:

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